THE UNIVERSITY OF SYDNEY

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T1-Group5

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Section 1: Concept and Design Rationale

1.1 Design Goal and Rationale

Our project reimagines the legendary Gandalf's staff from *The Lord of the Rings* as a real-world navigation companion. In Tolkien's stories, Gandalf's staff symbolises light, guidance, hope, and protection. We wanted to capture that same sense of "magic" through technology. The result is a smart walking staff that helps users explore places safely and confidently, while preserving the iconic presence of the original artifact.

The staff uses modern sensors and AI-driven interaction to guide users through their surroundings. It lights up and provides directional feedback using LEDs. This combination of sensing, context-awareness, and adaptive response makes it both a practical aid and a playful nod to fantasy.

From the technical perspective, the system integrates an ESP32-E microcontroller, a GPS module, a magnetometer, a button, and a NeoPixel ring for illumination. These components work together to detect the user's environment and provide real-time feedback. The enclosure is custom-designed and 3D printed to resemble Gandalf's staff, combining aesthetics, durability, and functionality.

Our goal was to explore how a mythical object could be reinterpreted through the principles of pervasive computing: embedding intelligence, awareness, and interaction into a familiar physical form. By blending creativity and engineering, the project turns a piece of fantasy into a tangible, interactive experience.

1.2 User Experience Overview

The user experience focuses on intuitive interaction through simple physical input and clear visual feedback. The design ensures that users can engage with the staff naturally, without relying on screens or external devices.

1.2.1 User Journey

When the user picks up the staff, it immediately powers on and adjusts to the environment. The button on the handle is the main form of user input. Each press represents the user's intent: the more times they press, the further away the suggested exploration location will be. This converts a simple gesture into a scalable command, keeping the experience direct and accessible.

1.2.2 Feedback and Interaction

The LED ring provides all feedback through light patterns that correspond to each stage of interaction:

- **Input Phase**: The LEDs glow yellow while the user presses the button, confirming that input is being registered.
- **Processing Phase**: The LEDs shift to a white glow or pulse while the system calculates the destination suggestion.
- **Navigation Phase**: Once the route is determined, the LEDs turn blue and illuminate in the direction the user should head toward.
- **Reaching Phase**: When the user approaches 50m to the destination, the LEDs flash yellow to indicate to the user that they are close to the destination.
- **Arrival Phase**: Upon reaching the destination, the LEDs light up in a rainbow color to indicate completion.

This flow of colours and animations gives continuous visual cues without requiring the user to check a screen or app, maintaining immersion and simplicity.

1.2.3 Context Adaptation

The staff uses sensor data to remain responsive to its surroundings. The magnetometer tracks orientation, and the GPS determines the user's coordinate position as well as providing data to the weather API which is used by the gemini AI in determining an appropriate location for navigation. Together, these enable dynamic direction feedback and adaptive lighting based on real-world conditions.

1.2.4 Accessibility and Intuitiveness

The interaction design emphasises one-handed operation and minimal cognitive load. The colour-coded LED feedback is instantly understandable through colors that match the real world in terms of status and meaning, and the single-button interface supports use by people with limited dexterity or technical experience. By combining tactile and visual elements, the staff delivers a clear and satisfying user experience that merges functionality with the project's magical inspiration.

Section 2: System Design and Implementation

2.1 System Design Overview

This section provides an overview of the system design, describing how the hardware, software, and enclosure work together to form the complete prototype. The goal is to show how each part contributes to the overall operation of the staff. Detailed Assembly Instructions are included in the Appendix; usage instructions are included in the User Manual.

The system integrates a set of sensors, a microcontroller, and feedback components to create an intelligent and responsive walking staff. At its core is the ESP32-E microcontroller, which collects input from the GPS module, magnetometer, and a button. The microcontroller processes this information and controls a NeoPixel LED ring to deliver visual feedback to the user.

The button plays a key role in user interaction. The system records each press, and the total number of presses determines how far the suggested location will be. The onboard AI module uses this input, along with the user's current GPS position, to suggest suitable nearby places to explore. This feature converts a simple physical input into an intuitive form of intent expression, enabling the system to interpret the user's level of adventure and adapt its suggestions accordingly.

The hardware is housed within a custom 3D-printed enclosure mounted on a 1.5 m wooden pole. Each electronic module is placed within its own compartment for easy assembly and protection. The internal wiring passes through the central section, maintaining a clean layout while ensuring reliable connectivity between all components.

The software, developed in the Arduino environment, coordinates communication between modules and interprets sensor data to determine suitable responses. The magnetometer and GPS module detect directional changes, and the LED ring displays guidance patterns.

Together, these elements form a cohesive system that combines sensing, interaction, and intelligence. The ESP32 acts as the control hub, the sensors provide environmental awareness, and the LEDs offer immediate visual feedback. The enclosure ties everything together in a design that feels natural to hold and use. The result is a smart staff that blends practical assistance with creative expression, embodying the project's central idea of pervasive computing: intelligence embedded seamlessly in everyday objects.

2.2 Hardware Implementation

2.2.1 Hardware Components and Specifications

The hardware design coherently integrated various sensing, processing, and output elements toward a contextual navigation system. Each module was selected concerning its compatibility with the ESP32 platform, low power consumption, and reliable communication interface. Table 1 contains the whole list of components and their relevant specifications.

Table 1. Hardware components and specifications

Component	Model / Type	Key Specifications	Function / Notes
Microcontroller	DFRobot FireBeetle 2 ESP32-E	Dual-core 240 MHz CPU, 4 MB Flash, Wi-Fi 2.4 GHz, Bluetooth 4.2 BLE, 3.3 V logic	Central processing unit; runs all sensor tasks, LED control, and API communications
GPS Module	u-blox NEO-6M	UART interface, 9600 bps, 2.5 m CEP accuracy, 50 channels	Provides latitude, longitude, speed, satellite count, and HDOP data
Magnetometer	Adafruit MMC56x3	3-axis magnetic field sensing, ±30 Gauss range, I^2C interface	Detects compass heading; used for orientation and target bearing alignment
LED Ring	WS2812B NeoPixel (16 LEDs)	Individually addressable RGB LEDs, 5 V input, 60 mA max per LED	Displays real-time navigation direction, proximity alerts, and visual feedback
Button Input	Tactile push button, GPIO14	Digital input, internal pull-up resistor	Used to trigger API queries and select search radius modes
Power Source	USB 5 V or Li-ion battery (3.7 V, 2000 mAh)	Current draw ~400 mA (active Wi-Fi + LED)	Provides portable power; suitable for short-term field testing
Connectivity	2.4 GHz Wi-Fi network	WPA2 secured access	Enables communication with Google Gemini, Weather, and Places APIs

2.2.2 Circuit Diagram and Connection Map

The following diagram demonstrates the circuit connections between each individual component with the main ESP32-E microcontroller.

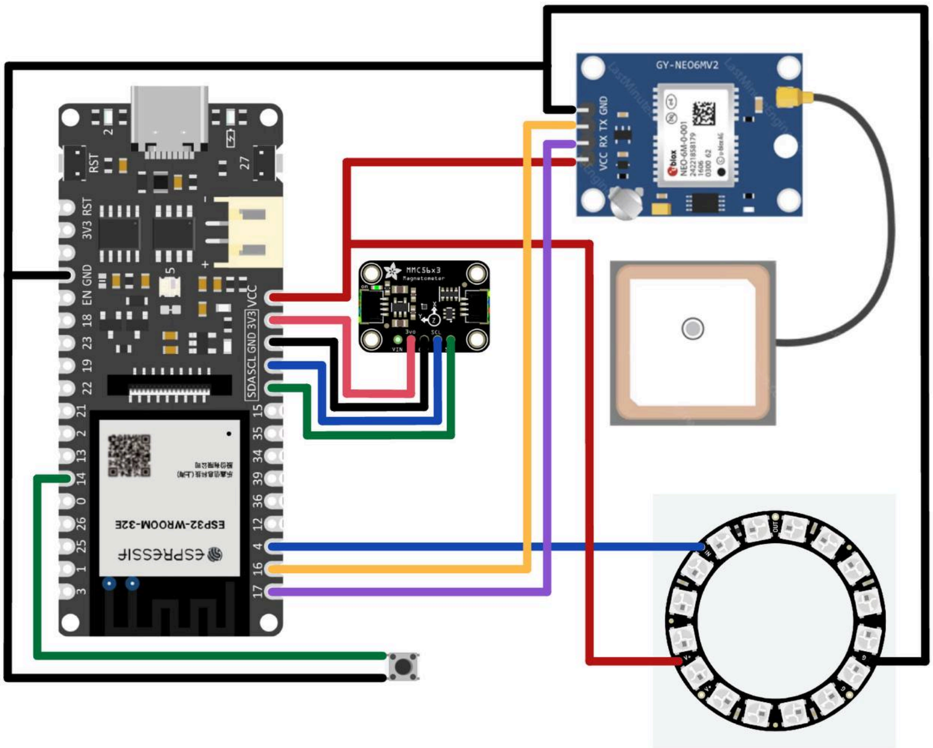

Fig 1. Circuit Diagram and Connection Map

2.2.3 Assembly and Wiring Process

Our prototype's assembly encompassed external structural construction and internal circuit integration, thus combining modularity with maintainability.

External Structure (Enclosure)

The enclosure was done using 3D printing technology. Each component was printed separately and finally put together to complete the structure. This enabled flexibility during the assembly stage and allowed easy disassembly for troubleshooting whenever problems occurred with the internal layout or circuitry.

Internal Wiring Layout

Prior to assembly, all wiring routes were carefully planned and labelled. The pre-layout maintained correct pin identification and accurate connections when integrating the various modules. This, in effect, made the wiring more organised with reduced time spent and minimized the chances of misconnection.

In practice, we partially assembled the enclosure first to protect sensitive components—especially the GPS module, which had suffered physical damage due to prolonged exposure during testing. Some wires had fractured, requiring replacement with spare modules. After completing the individual module testing, each component was integrated into its corresponding enclosure section. Once the internal circuit logic was fully developed and verified, all wiring and enclosure parts were assembled together to form the final prototype.

2.2.4 Power Setup and Battery Management

Based on all our existing modules, development boards, and batteries, we have compiled statistics on the power supply's specifications.

Table 2. Modules and Components Utilised in the System with Typical Current

Modules	Current (mA)	Instruction
ESP32 (FireBeetle 2 ESP32-E)	70	Main control chip (WiFi off, background operation)
u-blox NEO-6M GPS	35	Continuous tracking and positioning status
Micro SD Card Module	5	Occasional write operations, average ~ 5mA
MMC5603 Magnetometer	1	Continuous orientation measurement
NeoPixel Ring (16×WS2812)	80	Direction prompt mode illuminates 4 LEDs
Mini Push Button	0	Input button operation, negligible power consumption

This system is built on the DFRoot FireBeetle 2 ESP32-E microcontroller, which combines a u-blox NEO-6M GPS module, an Adafruit MMC5603 magnetometer, and a NeoPixel ring LED to create a multi sensor navigation device. All the modules are powered by a single-cell lithium poly battery at 3.7V / 1000 mah. From conservative estimates of the operating currents, the main ESP 32controller consumes about 70 mA, the GPS modules 35 mA, the magnetometer 1 mA, and the LED ring approximately 80 mA when four out of its 15 LEDs are lit. Hence, total current consumption is about 190 mA. **The system can be expected to run for roughly 5 hours under such conditions of continuous operation,** taking into account power conversion losses, hence meets short outdoor positioning and directional indication needs.

Power consumption Analysis reveals that the two major sinks of energy are the LED ring and the ESP 32-E, which account for nearly 80% of total power consumption. Under a condition where all 16 LEDs are turned on, total current may reach 410 mA, thus limiting its operation to about 2 hours. The introduction of Wi-Fi connectivity or increase in data writing frequency would bring about an average current of around 245 mA, which extends its operation to close to 4 hours. Therefore, some key strategies to extending battery life will be contending with the reasonable control of LED brightness and quantity, minimizing continuous communication, and optimising task execution frequency. Above all, there is an excellent compromise between performance and energy efficiency navigation and perception are guaranteed while the battery carries an excellent compromise in its capacity.

2.2.5 Hardware Integration Challenges and Solutions

During the build of our prototype, where we used the previous circuit diagram, we faced typical issues related to assembly quality system stability.

Enclosure Assembly Problems

The enclosure assembling was infeasible because the parts were printed successively. During the bonding, the part either broke or misalignment took place. The final prototype's structurally gross and aesthetic quality was performed poorly.

Unstable Wiring and Connection Problem

Ensuring the wiring fit securely and reliably into the enclosure was one of the major challenges faced in the hardware integration process. Due to the multicomponent nature of the design, the internal circuitry layout was prone to overcrowding.

Connections lost or wrongly connected with signal loss. In particular, considerable issues arose with the GPS module. Since indoor testing conditions applied, the GPS unit could not be guaranteed to pick up a signal all the time. Hence, system errors occurred in its integration. Unfortunately, these were vaguely indicated, and therefore diagnosing those became challenging.

Poor Soldering Quality

The initial soldering work was not that good. Some solder joints are weak and break with minor force impacts. Hence, such solder joints make the electrical connection unreliable in operations.

We will improve the final assemblies through further refinements in the assembly process and wiring layouts. The final prototype demonstrated sufficient functionality and robustness for the demonstrations. However, it still requires optimisation and refinement for commercial viability.

2.3 Software Implementation

2.3.1 APIs and Libraries

In this project, we put together several APIs and Arduino libraries to achieve navigation, environmental awareness, and visualization using the ESP32 platform. Google Gemini API was used for smart data processing and communication, while Google Places API was used to get coordinates and information about the places around. On top of that, we added a third-party Weather API for the live weather information about temperature and conditions. Since the Weather API is external, it has to be handled separately with an API key, while all other Google services are handled using a single Google Developer API key.

On the libraries side, we used ArduinoJson (bblanchon/ArduinoJson @ ^7.0.0) to manage and parse the JSON data received from the APIs, TinyGPSPlus (mikalhart/TinyGPSPlus @ ^1.1.0) to extract and interpret the GPS coordinates, FastLED (fastled/FastLED @ ^3.10.3) to control the LEDs as an indicator of the direction, and finally Adafruit MMC56x3 (adafruit/Adafruit MMC56x3 @ ^1.0.2) to read the magnetic heading from the magnetometer. With all these components, the ESP32 will determine its position and orientation and acquire local weather and place information to visually represent the navigation directions through dynamic LED lighting.

2.3.2 Data Flow Diagram

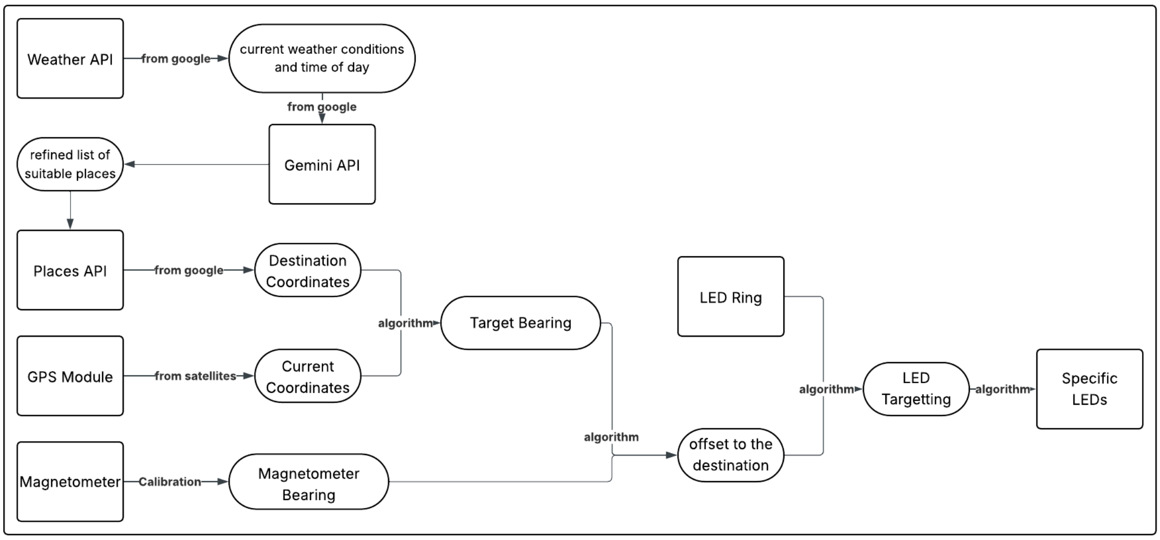

Fig 2. Data Flow Diagram of Navigation System

The diagram illustrates the overall system architecture and data flow of our ESP32-based navigation project, which integrates multiple APIs and sensor modules. The process begins with data acquisition from three main sources: Google APIs, satellite signals, and onboard sensors. The Weather API provides current weather conditions and the time of day, while the Google Places API returns a list of nearby locations and their coordinates. These two inputs are processed through the Google Gemini API, which refines the list of suitable destinations based on real-time weather and contextual factors. The selected destination coordinates are then passed to the navigation module.

Simultaneously, the GPS module continuously retrieves the device's current coordinates, and the magnetometer provides a calibrated bearing relative to magnetic north. An internal algorithm calculates the target bearing—the direction from the current position to the destination. This bearing is compared with the magnetometer's reading to determine the offset to the destination, which is then used to drive the LED Ring. The LED Targeting algorithm activates specific LEDs to visually indicate the correct direction toward the destination.

2.3.3 Pseudocode for Key Functions

```
// ---------------------------
// System Setup
// ---------------------------
setup():
    start Serial & WiFi connection
    init LED, GPS, and Magnetometer
    calibrate magnetometer (10s)
    store reference heading as baseline

// ---------------------------
// Main Loop
// ---------------------------
loop():
    if Gemini query not triggered:
        detect button presses for 5 seconds
        pressCount = number of presses (→ search radius)
        get GPS coordinates (latitude, longitude)

        // Fetch real-world data
        weatherCondition, localTime = fetchWeather(lat, lng)
        filteredPlaceTypes = queryGemini(weatherCondition, localTime)
        destLat, destLng = fetchNearbyPlace(filteredPlaceTypes, pressCount, lat, lng)

        set queryGeminiFlag = true
    else:
        // Navigation Mode
        if GPS fix valid:
            nav = computeNavigation(currentLat, currentLng, destLat, destLng)
            handleNavigationLED(nav)

            orientation = computeOrientation(currentLat, currentLng, destLat, destLng,
refHeading)
            displayOrientationLED(orientation)

// ---------------------------
// API & Sensor Functions
// ---------------------------

// Get weather info (Weather API)
fetchWeather(lat, lng):
    call WeatherAPI with coordinates
    return condition + local time

// Filter place types (Gemini API)
queryGemini(weatherCondition, localTime):
    send weather + time context to Gemini
```

```
    receive list of suitable place types (e.g., ["park", "museum"])

// Find nearby destinations (Places API)
fetchNearbyPlace(types, radius, lat, lng):
    query Google Places within radius
    return random destination coordinates

// Compute navigation details
computeNavigation(currentLat, currentLng, destLat, destLng):
    use Haversine formula → distance, bearing, ETA

// Compute relative orientation
computeOrientation(currentLat, currentLng, destLat, destLng, refHeading):
    compare magnetometer heading with target bearing
    return relative angle for LED indication

// LED feedback
displayOrientationLED(orientation):
    light up LED ring toward destination direction
```

2.3.4 Context-recognition Logic

The system's context-aware, dynamic behavior underlays many sources of environmental and spatial data. This means that the Weather API outputs the current weather and local time, which are then used as contextual parameters for the Google Gemini API. This, in turn, analyzes the information and returns a refined list of appropriate destination types as per the case (e.g., parks on sunny days or museums in case of rain). This is based on environmental context recognition , where the system adjusts its destination suggestions according to real-world situations.

Live data from the GPS module and magnetometer yield a continuously changing spatial context. The GPS module gives the current values for latitude and longitude, while the magnetometer measures the user's orientation relative to magnetic north. The navigation algorithm processes these inputs to yield the target bearing and the relative angle between the user's facing direction and the destination. The LED ring will then update dynamically to point to the destination, providing real-time directional guidance.

Proximity context is recognised through distance thresholds. When the user approaches within 100 meters of the target, the LED ring performs a yellow "breathing" LEDs; at 50 meters, a rainbow effect plays before the system stops. Such visual effects provide immediate context-sensitive feedback, which continues to change as the user approaches the goal.

2.3.5 Error Handling / Adaptation Features

The system is designed with numerous fault-tolerant, adaptive mechanisms that will be presented to show its stable operation under real-world conditions. The Wi-Fi connection routine attempts to connect to an appropriate network repeatedly until it acquires a valid network link, thus guaranteeing reliable access to Google and third-party APIs. The GPS task queries gps.location.isValid() for verification and continues requesting data until valid latitude and longitude values are obtained. This is done to prevent, as much as possible, any ill effects on navigation caused by poor or weak satellite signals.

The magnetometer module does an automatic 20-second calibration at startup, dynamically adjusting offsets and scale factors to match local conditions of magnetic interference. Thus, the integrity of heading detection is preserved at all locations. Therefore, every API call, such as Gemini, Places, and Weather, checks the HTTP response code and safely deserialises the JSON data with ArduinoJson. In case of any error or missing data, the diagnostic message is printed out on the serial monitor, which allows one to recover smoothly and debug easier.

Since these strategies are combined, the system has shown high robustness to network failure, sensor noise, and environmental changes. In other words, this is a robust embedded context-aware navigation platform that resiliently responds to user activities and changes in external conditions rather intelligently.

2.4 Enclosure Design

The staff is made up of a wooden pole and a 3D-printed enclosure. The wooden pole, measuring 25 mm in diameter and 1.5 m in length, was purchased early in the project. This dimension was then used as the basis for designing the enclosure, which forms the top part of the staff.

The enclosure was printed using the Bambu Lab X1-Carbon available at ThinkSpace. The design has undergone more than 10 iterations to reduce print time, minimise the risk of print failure, and simplify the assembly process. A complete list of all enclosure components is shown below:

Table 3. Enclosure Component List

Name	L x W x H (mm)	Purpose
Handle	30 × 30 × 100	Connect the enclosure to the wooden handle
Adjuster	44 × 44 × 20	Adjust the height of the overall assembly
Central	70 × 70.5 × 120	House the ESP32-E, the button, and the charging port
Mag	70 × 70 × 8	House the magnetometer
Neo	70 × 70 × 77	House the Neopixel Ring 16 LEDs
Light	70 × 70 × 50	Allow light to pass through
Connector	50 × 50 × 15	Allow the wire to go through without entanglement
Top	50 × 50 × 25	House the NEO-6

All parts were initially assembled in Tinkercad to make sure the wiring could be done easily. We also made small adjustments, such as creating a small hole in the connector so the wires can pass through properly.

2.4.1 Handle

The Handle connects the wooden pole (25 mm × 1.5 m) to the main enclosure. It features a 27 mm inner diameter for a snug fit, with sufficient tolerance to ensure smooth assembly. The part is glued in place for a secure connection and acts as the structural base supporting the upper enclosure and electronics.

The design proved to be stable, with the main revision being an increase in contact length from 50 mm to 80 mm in the final version, along with slight adjustments to the inner diameter. This change enhanced overall strength and improved the reliability of the connection between the pole and enclosure.

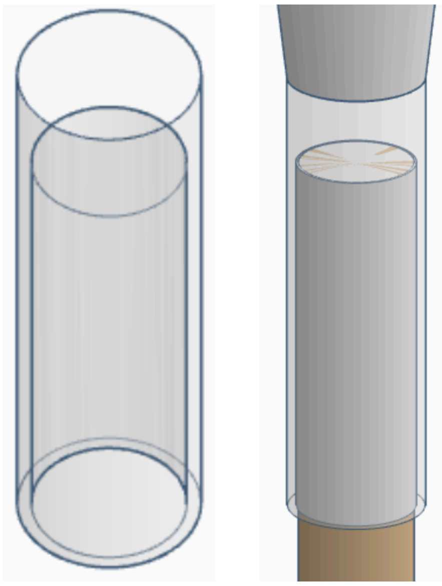

Fig 3. 3D model of the Handle (left) and its assembly with the wooden pole (right), showing how the enclosure fits securely onto the pole connection.

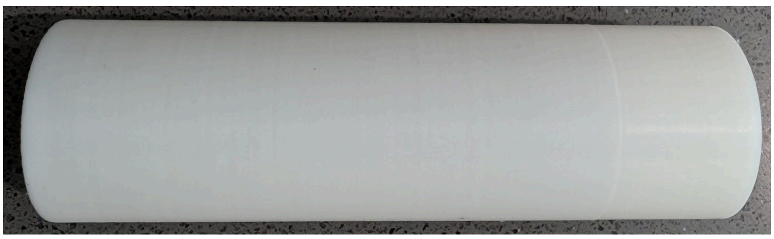

Fig 4. The printed model of the Handle

2.4.2 Adjuster

The Adjuster connects the Handle and the Central section, keeping everything aligned and stable. The total height of the staff ends up slightly taller than the average adult height in Australia. This proportion gives the staff a balanced look and a natural feel when held upright, ensuring it appears well-sized for everyday use while maintaining a clear, functional structure.

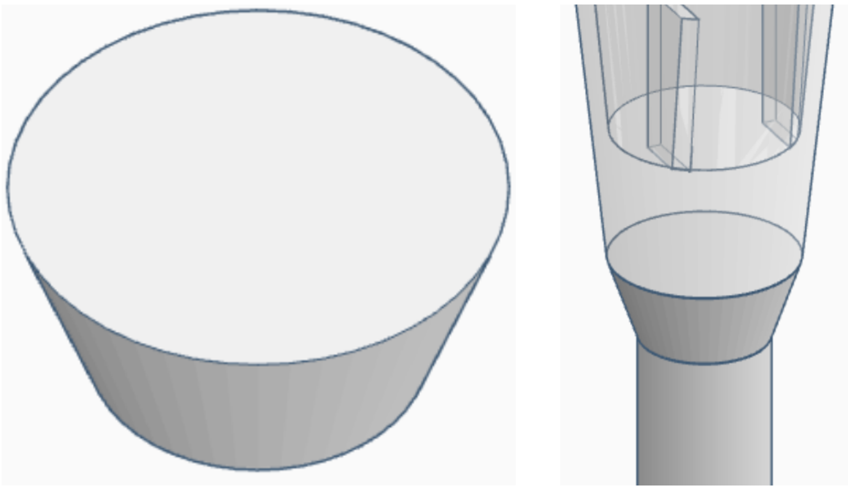

Fig 5. 3D model of the Adjuster (left) and its assembly with the Handle and Central (right), showing its position between the two parts.

Fig 6. The printed model of the Adjuster

2.4.3 Central

The Central is the main housing unit that contains the ESP32-E microcontroller, the button and the charging port. This part required the most detailed internal design to manage the placement of components and wiring efficiently.

The ESP32-E is positioned between two supporting columns and secured with adhesive. The charging port utilises a 90-degree USB-C adapter, providing easy access for charging without compromising the enclosure's structure.

The design of the Central went through several iterations. During this process, the magnetometer was moved to the Mag, and the battery was moved to the Neo, as shown in the figure below. We also remove the microphone and its housing slot from the final implementation. These changes simplified the Central's structure, made it easier to print, and helped shorten the internal wiring.

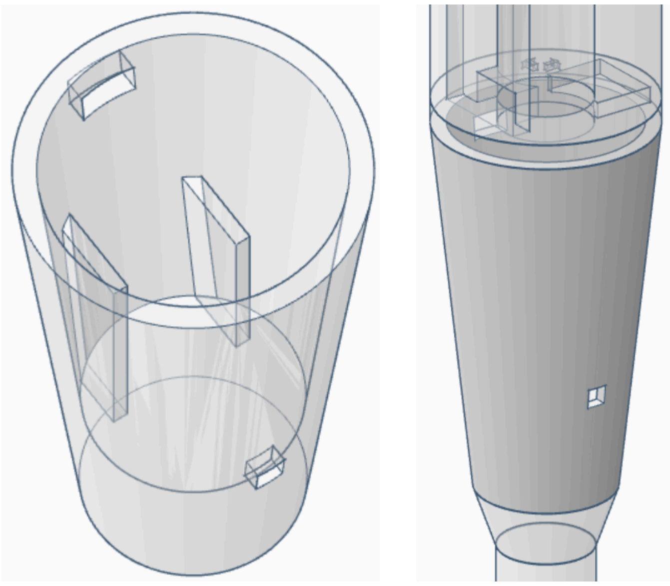

Fig 7. 3D model of the Central (left) and its assembly with the Mag and Neo (right)

Fig 8. The printed model of the Central

2.4.4 Mag

This section houses the magnetometer. After several design iterations, the final version streamlines the printing process by incorporating a dedicated slot at the top for inserting the magnetometer, instead of in the middle. This change improves print efficiency and minimises the risk of structural collapse. The surface is also etched with arrows to ensure correct magnetometer orientation during assembly.

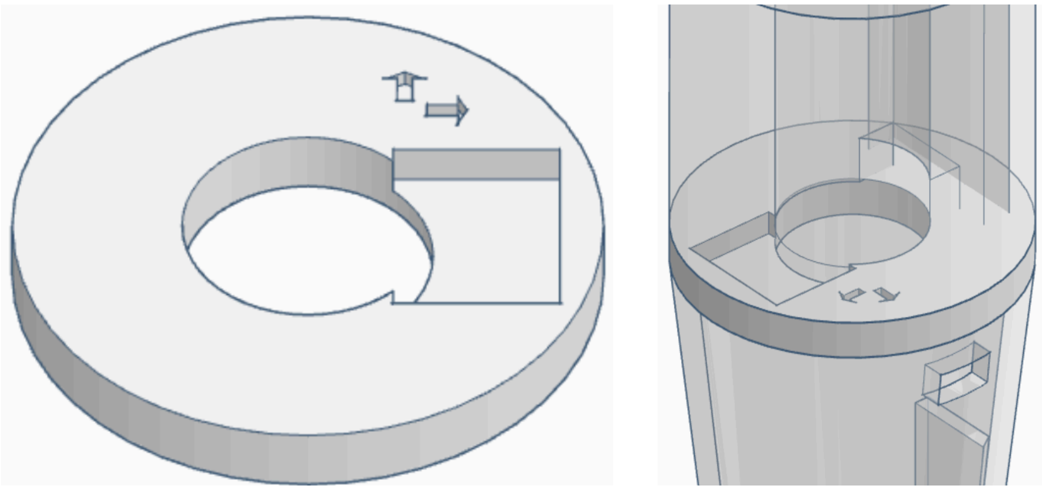

Fig 9 3D model of the Mag (left) and its assembly with the Central and Neo (right)

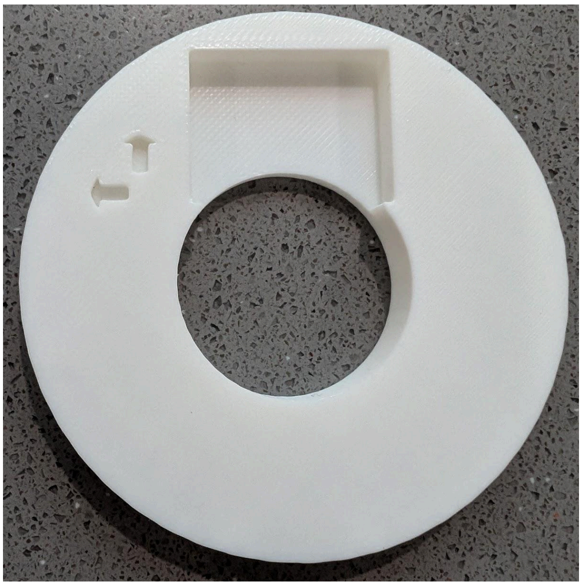

Fig 10. The printed model of the Mag

2.4.5Neo

This part houses the NeoPixel LED ring at the top, which is held securely in place by four supporting stands that provide sufficient clearance for the wiring to connect between them. The battery is inserted and fastened from the bottom of the part, positioned to ensure that the distance between the battery and the ESP32-E underneath remains minimal. A middle opening runs through the structure, allowing the wiring to pass and connect seamlessly to the Central.

The final design maintains a clean and organised internal layout while simplifying assembly and minimising the overall footprint. Through several design iterations, the geometry and component alignment were refined to improve reliability, accessibility, and ease of integration within the broader system.

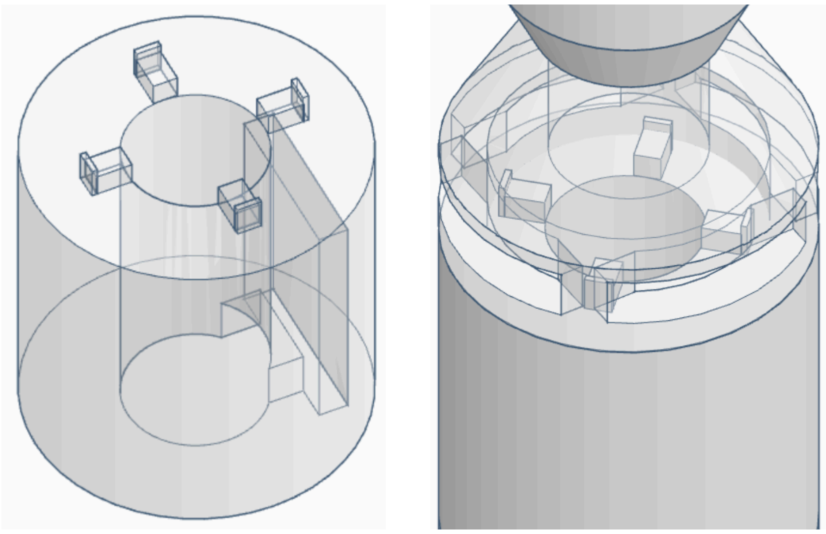

Fig 11. 3D model of the Neo (left) and its assembly with the Light (right)

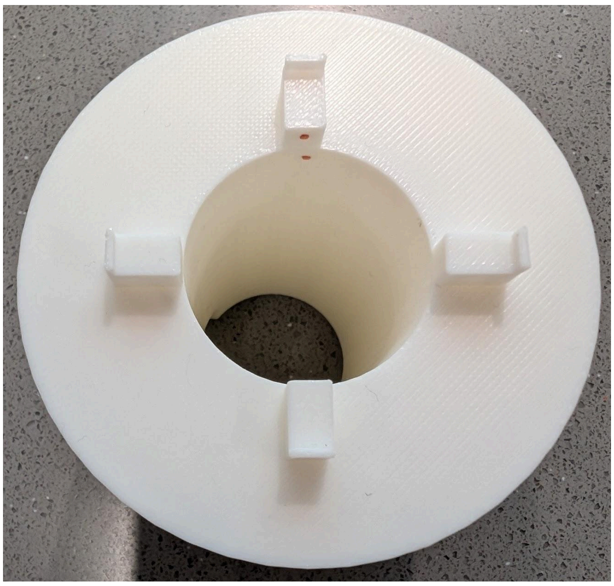

Fig 12. The printed model of the Neo

2.4.6 Light

This part functions as a light diffuser, allowing illumination from the NeoPixel ring mounted on the Neo to pass through clearly. Its geometry directs the emitted light outward, achieving efficient diffusion while maintaining the structural strength and dimensional stability required for assembly. The four openings positioned around the base allow light to escape horizontally, providing users with a coarse yet intuitive sense of orientation during navigation. When the NeoPixel ring displays different patterns or colours, these side windows enable users to visually gauge direction or system status without relying on a full visual display.

The design of this part has undergone several iterations to improve printability and reliability. The initial version featured complex hole shapes that required extensive tree supports, leading to longer print times and a higher risk of print failure. The final version adopts a simpler and cleaner geometry that reduces support material, shortens fabrication time, and increases the overall success rate of printing.

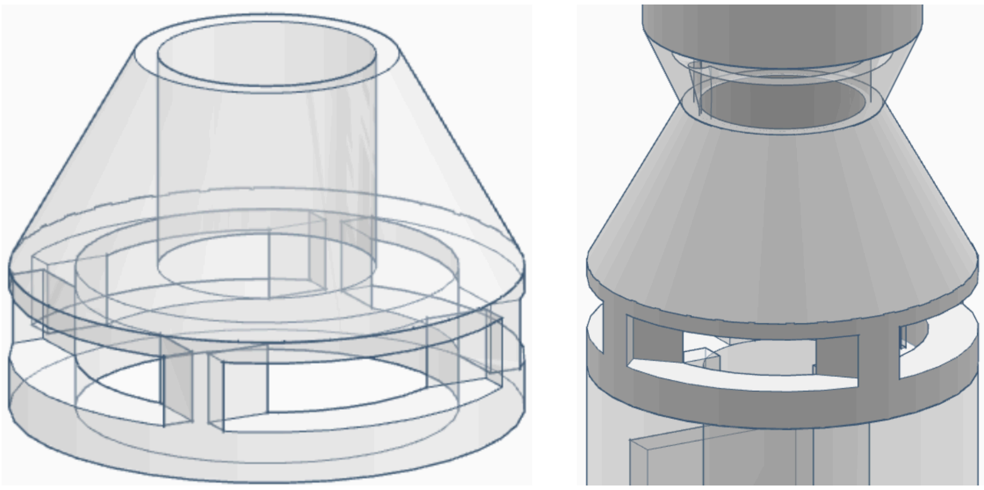

Fig 13. 3D model of the Light (left) and its assembly with the Neo and Connector (right)

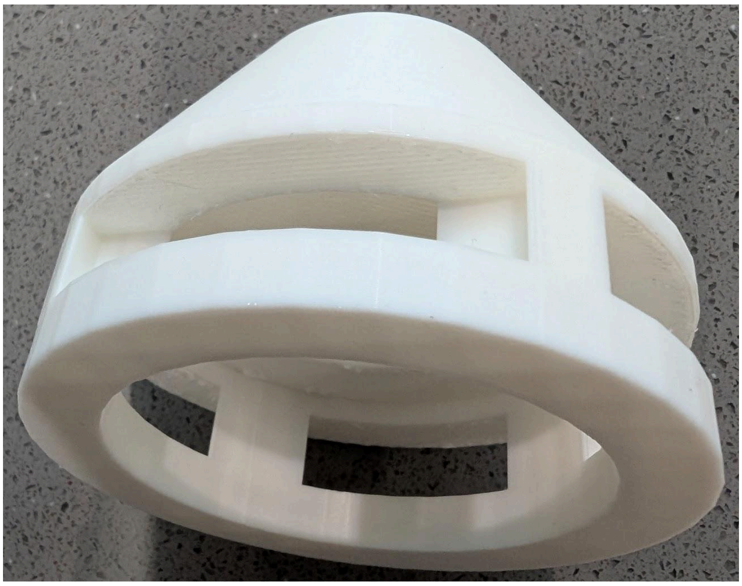

Fig 14. The printed model of the Light

2.4.7 Connector

The Connector acts as an interface between the Top and the Light, providing a clean and organised path for the wiring. A through-hole was added to allow the wire to pass through the assembly. The design remained simple and functional throughout development, requiring only minor adjustments while maintaining reliable printability and consistent integration with the other components.

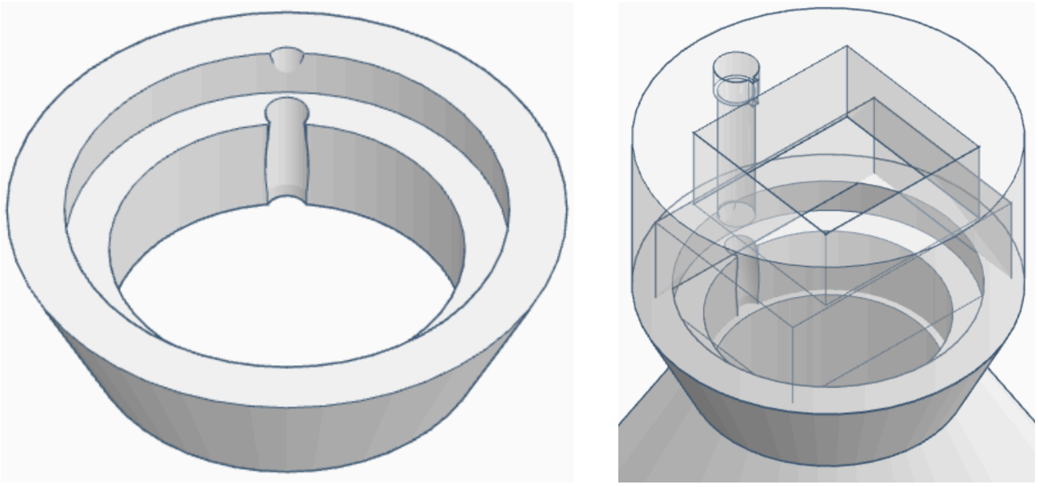

Fig 15. 3D model of the Connector (left) and its assembly with the Top (right)

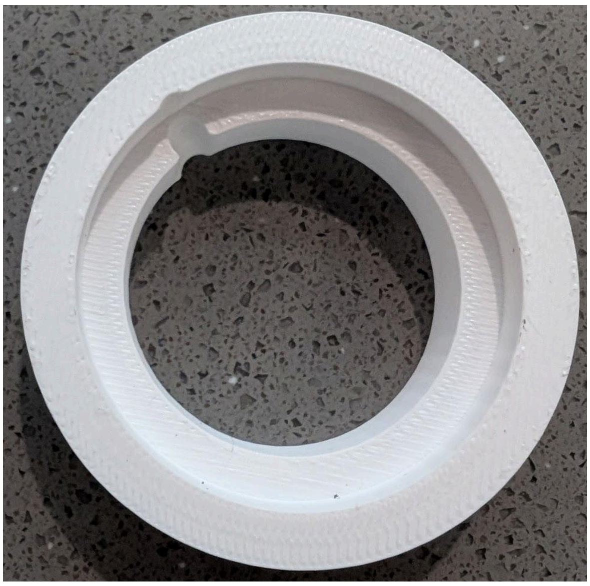

Fig 16. The printed model of the Connector

2.4.8 Top

The Top houses the NEO-6 GPS module. The NEO-6 module is positioned with a clear view of the sky so it can receive GPS signals reliably. The Top connects to the Connector at the bottom to complete the upper assembly.

Several design iterations were made to fine-tune the dimensions, achieving a precise fit for the NEO-6 module. The NEO-6 has two subcomponents connected by wiring: the upper part sits on top of the printed structure, while the lower part is inserted from underneath, creating a stable and tidy assembly.

Note: We made a decision to remove the light sensor, so the Top only contains NEO-6 as described in the main text.

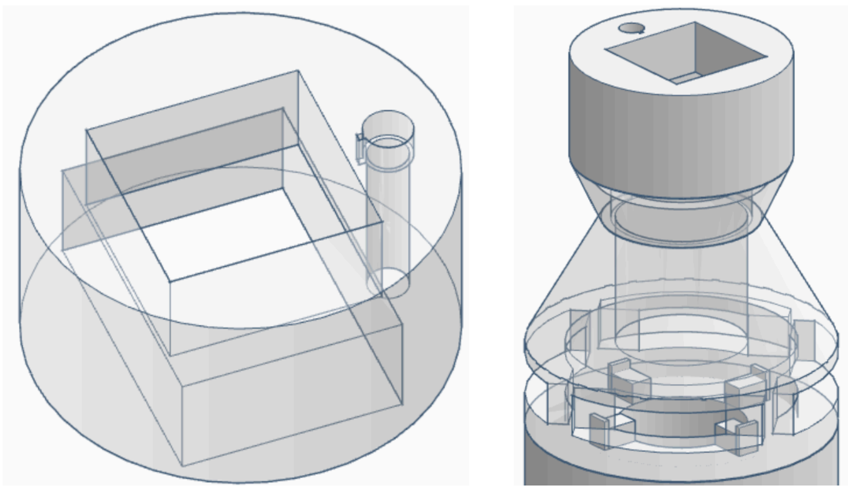

Fig 17. 3D model of the Top (left) and its assembly with the Connector (right)

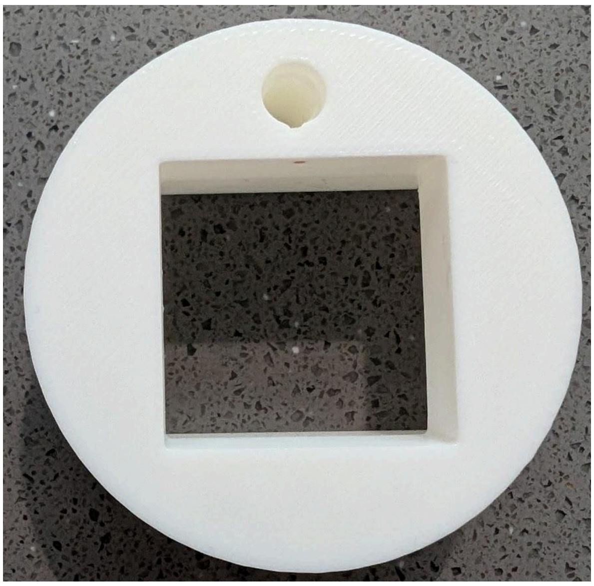

Fig 18. The printed model of the Top

2.5 Integration & Testing

2.5.1 Components for Testing

The integration and testing phase was performed on both hardware and software, which together comprise the holistic pervasive navigation system. As the system's flow diagram shows, this project involves several different, but highly interlinked, modules, with each module implementing one specific layer of data acquisition, processing, or output.

Table 4. Expected Testing Outcomes from Modules/APIs

Modules/API	**Expected Testing Outcomes**
Weather API	Returns real-time weather conditions and local time. This data provides contextual input that influences the choice of destination type.
Gemini API	processes the weather and time information, running AI filtering to produce a shortlist of appropriate place categories.
Places API	Categories returned by the Gemini API are specified in latitudinal and longitudinal terms for suggested destinations within a given radius.
GPS Module (u-blox NEO-6M)	record the current position in a continuous loop, extracting coordinates, speed, satellite count, and HDOP from the satellites to provide input for navigation calculations.
Magnetometer (Adafruit MMC56x3)	identifies the magnetic field and, when calibrated, gives heading information so the user can tell their orientation concerning true north.
Target-Bearing Algorithm	uses the current position from GPS and the destination coordinates to find the great-circle bearing (direction) via the Haversine formula.
LED Ring (WS2812B via FastLED)	gives a computed offset between the user's facing direction and the target bearing, turning on some of the LEDs to indicate the direction.
LED Targeting Algorithm	Interprets navigation results to select which quadrant or LED group should illuminate for real-time directional feedback. Functional testing also implemented proximity-based light patterns, such as yellow breathing within 100 m and rainbow animation within 50 m.

2.5.2 Testing Method

This is how the entire system was made operational, accurate, and stable by using different combinations of unit, integration, and field testing. Special attention was paid to data flow and performance under observable and real conditions of combined testing of various sensors, APIs, and the LED output module.

Unit Testing

Checking independent operation for every unit was the very first step in testing. The validity of the GPS module was checked based on raw NMEA output. Latitude and longitude must indicate the correct position, speed, and number of satellites achieving success. To test the magnetometer heading consistency, it has to be calibrated with variations in position and orientation. The LED ring was checked for color, brightness, and its response to the given directional command. Sample queries testing was done by invoking the Weather, Gemini, and Places APIs to check authentication and responses.

Integration Testing

Once individual modules were stable, hardware and software were integrated to create a complete data pipeline from environmental input to LED output. At this point, the system's ability to process the sequential API calls was tested—Weather, Gemini, and Places—as well as the capability for live synchronisation with GPS and magnetometer readings. Serial logs monitored timing and data integrity to ensure bearings, distances, and LED indications were consistent throughout the real-time execution.

Field Testing

Real-life performance was tested in open outdoor environments. The ESP32 system was provided with USB power and Wi-Fi connectivity to access the API. The live position information was provided with the GPS module obtaining satellite data, whereas the magnetometer continuously updated the orientation. In accordance with the movements and rotations of the tester, information was collected regarding the ring of LEDs activated and in what direction. Simulation for button pressing tried out different search radii and context-based API queries for user interaction validation.

Performance Monitoring

Latency and responsiveness of the system were determined in this phase. The time delay in GPS updates LED activation was recorded. On average, it remained below a second. In addition, query times on the API and Wi-Fi reconnection stability were logged to assess communicating efficiency. In general, the approach to system testing has been such that the integrated system was expected to function reliably, exhibit dynamic responses to user movement, and maintain a level of accuracy under variable ambient conditions.

2.5.3 Observations and Measured Outcomes

Testing concluded that the proposed system could integrate hardware sensors, cloud APIs, and LED visual output. The GPS module performed well outdoors, too, with continuous, valid readings from 6 to 8 satellites on average and an HDOP of about 1.9, which approximates positional accuracy in the range of 2 to 3 meters. The raw NMEA data show evidence of consistent tracking and the reporting of speeds for this sample output:

*$GPGGA,130201.00,3348.27100,S,15110.74621,E,1,07,1.51,123.3,M,19.5,M,,*46*
*$GPRMC,130202.00,A,3348.27102,S,15110.74618,E,0.105,,250925,,,A*60*
*$GPVTG,,T,,M,0.105,N,0.195,K,A*2A*

The navigation and orientation algorithms processed this, making understandable summaries of the system's performance that reflected reality in movement. Hence, it was processed to provide travel distance, bearing, speed, and signal quality over time. Some results filling in those parameters are given here:

[10s] moved 11.5m, bearing 278.7° (W), speed~0.82m/s, hdop=1.9, sats=7
[status] fix=OK, sats=6, hdop=1.9, speed=1.21 m/s, course=263.1°
[10s] moved 8.2m, bearing 166.1° (SSE), speed~0.82m/s, hdop=1.9, sats=6

These results suggest that the navigation and bearing algorithms were properly implemented, as evidenced by the estimated travel direction and movement rate. The computed distance and bearing values validated the correctness of the Haversine concept in distance computation relative to physical movement, and the orientation computation using the magnetometer data was appropriate.

The output was in direct relation to issues of orientation and proximity throughout the testing. The user would rotate the device, and suddenly, the LED ring would update, showing the direction to the destination. The proximity-activated animations all worked correctly:

- At 100 metres, a yellow "breathing" animation of the LEDs played twice, indicating the approach.
- At a distance of 50 metres, a rainbow animation was activated for five seconds before the system shut down, having reached the destination.

On system responsiveness, the average delay in sensor updates and LED output was rounded up to less than one second, thus confirming almost real-time operation. The API calls for Weather, Gemini, and Places were successful within 2-4 seconds, depending on network conditions. There were no signs of instability in the systems over long periods, nor were there any signs of system crashes or communication interruptions. The integrated prototype was tested and found to perform accurately and promptly at a realistic level. It effectively integrated environmental awareness, spatial orientation, and visual feedback, thereby confirming the feasibility of the design and its projected function as a context-aware navigation assistant.

2.5.4 Prototype Photos

The final integrated prototype was to combine all hardware components in a small, portable system using an ESP32 FireBeetle development board. A breadboard and jumper wires were used here to connect the modules and make testing and debugging more accessible. Below are pictures showing the hardware setup, wiring layout, and real-life testing setup.

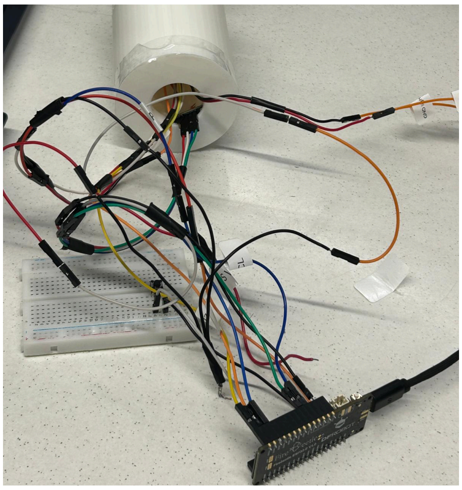

Fig 19. Complete System Integration

The developed ESP32 board is fitted with a u-blox NEO-6M GPS module, an Adafruit MMC56x3 magnetometer, and a WS2812B LED ring. The GPS module communicates through UART (pins 16 and 17), while the magnetometer uses the I^2C interface. The LED ring is powered and controlled using the FastLED library, which links to one digital output pin. Thus, the compact wiring layout allows modular testing and easy access for calibration.

Fig 20. Calibration and Magnetometer Testing

For the magnetometer calibration process, the system commanded a steady green light on the LED ring, indicating that the calibration was underway. The device was rotated slowly for about ten seconds, during which time the calibration was completed automatically, and the LED output stopped. This led to accurate magnetic bearing measurements.

2.5.5 Trade-offs

During system integration and testing, some trade-offs were found between accuracy, responsiveness, power efficiency, and environmental adaptability. One of the most challenging elements was the magnetometer. The Adafruit MMC56x3 sensor continuously updates heading data. However, once calibrated, this was very sensitive to nearby metal objects, electromagnetic interference, and small changes in tilt angle. Even post-calibration, subtle directional drift was evident, which occasionally spoiled the LED orientation accuracy. In order to mitigate this, the system employed an averaged heading filter with periodic recalibration; however, this slightly reduced responsiveness. A trade-off here was thus between smooth and stable direction readings and real-time responsiveness for LED updates. Beyond the magnetometer, other trade-offs pervade system design issues. The GPS module offered good positional accuracy with severe penalties regarding outdoor requirement-only usability. Indoor usage or heavy interference of this module would be immediately negated by such a choice. Continuous Wi-Fi and API communication increases the contextual intelligence of the system, offering data from Gemini, Weather, and Places, but this is offset by increased power consumption and dependence on network coverage. The LED ring offers bright, clear, and intuitive feedback but draws considerable currents when animating, mainly if the full brightness or color transitions are in play. Balancing these requires task scheduling and an update frequency optimally determined to reduce power draw while maintaining an uncompromised user experience. Again, the final prototype of this investigation offered a practically compromise model between performance, stability, and energy efficiency. Further development will require finer calibration of the sensors and compensation for magnetic interferences to make the product commercial or research-grade.

Section 3: Evaluation and Reflection

3.1 Results

3.1.1 GPS Modules

The project has come to yield several satisfactory and expected results after several cycles of development and tests, especially concerning the GPS module's development. Physical positioning of the system was achieved by developing and integrating the U-blox NEO-6M Compatible GPS Module and the FireBeetle ESP32-E development board. According to the user manual for the GPS module, the typical horizontal accuracy in Autonomous mode is about 2.5 meters CEP. Results from repeated testing match expectations and indicate that the system can adequately use this module as part of the final product. Throughout the testing and development phases, the GPS module in our system was perfectly functional. It delivered continuous, plentiful NMEA data streams used to validate signal stability, calculate navigation parameters, and perform key functions such as distance estimation, heading direction, and speed calculation. The data points facilitated quantitative analysis and corresponding algorithm design, transforming simple satellite information into significant movement metrics. As such, the enhancement of our understanding of the system's spatial behavior was more profound, and this provided an excellent basis for further development, such as navigation logic and LED direction indication.

Raw Data Acquisition

Below is an excerpt of the raw GPS output in standard **NMEA 0183** format, collected during live testing:

*$GPGGA,130201.00,3348.27100,S,15110.74621,E,1,07,1.51,123.3,M,19.5,M,,*46*
*$GPGSA,A,3,26,28,31,10,32,02,25,,,,,,2.27,1.51,1.69*07*
*$GPGSV,3,1,11,01,54,260,19,02,48,317,32,03,35,222,22,04,26,272,*7D*
*$GPGLL,3348.27100,S,15110.74621,E,130201.00,A,A*7F*
*$GPRMC,130202.00,A,3348.27102,S,15110.74618,E,0.105,,250925,,,A*60*
*$GPVTG,,T,,M,0.105,N,0.195,K,A*2A*

Each sentence encodes specific information:

- $GPGGA provides the main fix data: time (13:02:01 UTC), position (latitude 33°48.271′S, longitude 151°10.746′E), fix quality (1 = valid fix), number of satellites (07), horizontal dilution of precision (HDOP=1.51), and altitude (123.3 m).
- $GPGSA lists satellites used and overall accuracy (position dilution of precision values).
- $GPGSV gives detailed information on each visible satellite, including its signal strength (SNR) and elevation.
- $GPRMC records recommended minimum navigation data: valid status, position, speed over ground (0.105 knots), and course.
- $GPVTG converts speed into knots and kilometres per hour, confirming velocity consistency.

Processed and Quantified Data

After parsing and filtering through the **TinyGPSPlus** library, we transformed the raw NMEA outputs into human-readable summaries and quantitative movement reports, such as:

[10s] moved 11.5m, bearing 278.7° (W), speed~0.82m/s, hdop=100.0, sats=7
[status] fix=OK, sats=6, hdop=1.9, speed=1.21 m/s, course=263.1°
[10s] moved 8.2m, bearing 166.1° (SSE), speed~0.82m/s, hdop=1.9, sats=6

Each line represents a **10-second sampling interval** where:

- **Distance moved (m)** was calculated using the **Haversine formula**, comparing consecutive GPS coordinates.
- **Bearing (°)** indicates the direction of movement relative to the true north.
- **Speed (m/s)** was derived from successive coordinate differences and timestamp intervals.
- **HDOP** (horizontal dilution of precision) and **satellite count** serve as quality indicators for positioning accuracy.

After processing the data, the user's movement was projected onto the 3D space, and the navigation parameters were verified. Moreover, during the trip, the direction of travel, walking speed, and proximity could be detected in real time, which provided essential information for the LED guidance system. Hence, the GPS module was very stable and accurate, which laid the quantitative foundation for the navigation and context-based recognition features of the system.

3.1.2 Magnetometer

The magnetometer module went through multiple rounds of testing before it could be integrated into the overall circuit connections. Its purpose was to serve as the main source of directional awareness for the system through the detection of the Earth's magnetic fields and computing the raw data into headings that would be later used by the LED lights to guide the user towards the destination, along with compensating for the orientation of the staff. However, the implementation and testing phases highlighted the flaws and limitations of the magnetometer module.

The magnetometer readings from the initial testing phase were highly unstable and inaccurate, where even rotating it in a singular axis could not achieve usable results. The cause of this was later identified as the susceptibility of the module to external magnetic influences such as close proximity to other metallic objects and electrical components. Its sensitivity and the difficulty of obtaining even relatively accurate readings hindered the project progress in creating a responsive system. To mitigate the external factors effect on the magnetometer, calibration of the module became a crucial priority in the development of the prototype. However, calibration software for our particular magnetometer module was outdated and no longer available for download. Thus, only simple calibration using basic example sketches from the Arduino libraries could be utilised, which provided some correction for soft and hard iron calibration.

Additionally, the readings obtained post calibration were further simplified into one of the primary cardinal directions: North, South, East or West. This improved the usability and reliability of the magnetometer which allowed it to fulfill its intended purpose in our navigation system, albeit with reduced directional precision.

3.1.3 LED Ring

The LED module is the final and most complex stage in the realization of the system since it finally presents visual feedback to the user intuitively concerning computed data, sensor readings, and algorithmic logic. All contextual recognition, environmental inputs, and navigation computations have been funneled into this LED system: in short, it is the primary output interface of the project.

According to the system flow diagram, this module realises an end-of-chain processor within a multi-tier data flow architecture. From an interfaced Weather API, a triangulated Gemini API, a Places API, a suitable GPS module, and a Magnetometer, the system calculates the target bearing and relative offset between the user's orientation and destination direction. Calculated parameters are then fed through an LED targeting algorithm, whereby the respective LEDs are illuminated on the ring to visually represent the direction to the destination. The LED ring thus becomes a real-time directional indicator: The front LEDs light up in blue when the user faces the destination. Should the user turn, side LEDs light up to direct a person back toward the correct orientation.

Furthermore, as one approaches the target, contextual colour and animation cues gradually give feedback- with yellow breathing effects within 100 m, rainbow animations triggering within 50 m, and finally, the LEDs turn off once the target is reached. This multi-layered visual feedback mechanism reduces complex sensor data to an immediate human-readable signal. Beyond simply providing indications, it manifests the integrated layers of computation in environmental adaptation (weather-based destination choice through Gemini + Weather API), spatial orientation (magnetometer and GPS fusion for target bearing), and human interaction (LED-based intuitive guidance).

3.2 Reflection

3.2.1 Creativity and Critical Thinking

This project showcases a lot of creativity and critical thinking because it brings together hardware sensing, cloud intelligence, and contextual decision-making. Instead of following the normal path of sensor-to-display, we built an adaptive, multi-layer system that interprets environmental and situational data to steer user interaction in a meaningful way. This is really creative when orchestrating real-time weather conditions, Google's contextual knowledge through Gemini, and live GPS/magnetometer data into one coherent decision-making process.

However, the most compelling piece of evidence of critical thinking came when considering data fusion. Instead of separating the APIs or sensors, we had to think about how the various data types complement one another. So, for example, weather and time would affect the kinds of places a user might choose, so these variables were used to refine place recommendations via the Gemini model before querying the Places API. We also critically considered how heading and bearing could be mathematically aligned, even developing algorithms for calculating the relative offset between the user's facing direction and the destination to ensure accurate real-world orientation feedback from the LED ring.

Furthermore, design implied continuous problem-solving and iteration with regards to LED mapping, bearing thresholds, data responsiveness with visible clarity- all relating to design choice so as not to clutter the user's mind. From proximity feedback color coding to effective API queries, everything has been driven by a logical analysis and reflection of the user experience. This indicates quite well our ability to think critically about the technical and human factors in implementing pervasive computing. Overall, the critical thinking in this project is reflected in crafting the limitations, data validation, algorithm refinement for robustness and usability, while the genuine spark of creativity lies at the intersection of AI-driven, contextual reasoning with tangible and embedded outputs.

3.2.2 Learning about Pervasive Computing

This project deepened our understanding of the principles of pervasive computing—embedded computation within everyday environments to support context-aware interaction—in a much more pragmatic way. We learned how to combine sensors, APIs, and cloud intelligence in a system that perceives its environment, interprets data from heterogeneous sources, and acts. An ESP32 was the single, central logic unit of the system, and, triggered by inputs from the GPS, a magnetometer, and several Google APIs, it built an autonomous system of decision-making. Through such a process, we learned the fundamental nature of pervasive computing: intelligent, continuous, and unobtrusive device responses to changes in real-world conditions.

That said, pervasive computing isn't just about devices being interconnected but rather one step above: a system that understands context and acts on it without any direct intervention by the user. Making automatic changes in recommendations and visual feedback based on environmental parameters such as change of weather, time of day, and orientation in space is a characteristic of our project. In developing this system, we understood the challenges regarding real-time sensing, data synchronization, and low-latency feedback in embedded settings. Most importantly, we learned to strike a balance between the level of complexity involving advanced technologies and that of usability. That is, how one would go about possessing a device that feels responsive, customized, and human-centric, which embodies the very essence of pervasive computing.

3.2.3 Improvements and Future Expansion

While the project met its primary objectives, there are many areas of improvement or extension concerning performance and usability. The guidance algorithm uses LED to improve continuity in directional transitions with real-time high angular accuracy, especially in a slow user walk or fast orientation change. Improvements in energy efficiency, GPS signal stability, and magnetometer calibration make the device suitable for long-term outdoor use. On the software side, communication with the cloud can be optimised in an operation through deferred API handling and caching, which can decrease latency. Further enhancement includes the integration of voice or haptic feedback, Bluetooth connectivity, and smartphone integration for more active user interaction. Incorporating machine learning could enable the system to learn user behavior, dynamically suggest destinations, and adapt the guidance strategy over time, thus converting the prototype into a robust context-aware navigation assistant for research or commercial purposes.

3.2.4 From Prototype to Real Product

Further refinements on this prototype may let it grow into either a commercial or a research-grade navigation device. This actual implementation demonstrates essential real-life functionality: context-aware place recommendation, direction recognition, and real-time LED guidance. Further improvements in hardware compactness, weather resistance, and power management will make it a portable outdoor companion for activities such as hiking, tourism, or even accessible support for visually impaired persons. Integration of the ESP32 system with a mobile application would allow users to set preferences, view navigation history, and update the firmware over the air. From a research perspective, this project could be a testbed for human-computer interaction in pervasive computing, where contextual feedback—be it visual, auditory, or tactile—would help with perception and navigation efficiency. Indeed, the transformation of this prototype into a fully-fledged product would significantly broaden its usability and offer valuable empirical feedback on the real-life use of pervasive computing approaches.

3.2.5 Reflection on Teamwork and Time Management

This project called for solid teamwork and time management. Each member brought unique strengths, from hardware integration and code optimisation to API configuration and data analysis, which together enabled smooth progress through the development stages. We faced several challenges along the way, such as synchronising asynchronous sensor data, debugging concurrent FreeRTOS tasks, and aligning hardware testing with evolving software builds.

To handle this effectively, we adopted a modular workflow, assigning each subsystem its own responsibilities so it could be developed and tested independently before integration. These included the GPS, magnetometer, LED, and API modules. Regular meetings and consistent Git version control helped merge our work smoothly, avoid conflicts, and maintain a clear version history across the codebase.

Despite time constraints and occasional technical setbacks, the team showed strong communication, adaptability, and focus. The experience emphasized not just technical skill but also the importance of coordination, accountability, and iterative problem-solving in achieving a shared engineering objective.

3.3 Conclusion

This project interprets the iconic Gandalf staff as a contemporary navigation companion, combining fantasy with pervasive computing. The symbolic meanings of light, guidance, and protection that were inherited from the prototype took shape and breath into mythical features turned into a functional intelligent system. An ESP32-E microcontroller, GPS, magnetometer, and NeoPixel LED ring were embedded in the staff. It could sense its environment, automatically illuminate, and give real-time directional feedback. A perfect example, therefore, of pervasive computing that has incorporated sensing, context-awareness, and adaptive interaction into thematic design- making technology invisible and intuitive at the same time. Through storytelling and precision engineering, this project has shown how myth could exist side by side with modern computation to transmute a legendary artifact into an interactive, context-sensitive device and thus bring "magic" into the real world.

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Appendices

Appendix A: Self-Assessment and Declaration

We assessed our project against the official rubric and believe it meets the **High Distinction** standard, demonstrating mastery across all aspects of design, implementation, documentation, and presentation.

Prototype
Our prototype provides a novel and elegant response to the brief. It integrates an ESP32 microcontroller with multiple sensors and feedback, offering adaptive behaviour based on environmental context. The 3D-printed enclosure was fully custom-designed to accommodate user interaction. Component selection and circuit design were done creatively, resulting in a compact, reliable system suitable for real-world use.

Documentation
The documentation is polished and comprehensive, including complete circuit diagrams, component lists, and commented source code. The design rationale articulates both technical and user-centred reasoning, showing how feedback from teaching staff informed key iterations. The report also references the books and movies to justify design decisions. Each section was collaboratively written, and a clear user manual ensures replicability.

Teamwork and Process
The team maintained strong collaboration throughout the semester. Responsibilities were clearly divided, and regular progress reports were delivered on time. Peer feedback and tutor consultations were integrated into the design cycle. This consistent reflection and adaptation demonstrate excellent teamwork and process management skills.

Presentation
The final showcase presentation was polished and well-structured. The team used supporting materials and a video to communicate functionality and purpose clearly. The demonstration ran smoothly, and all members confidently explained technical and design aspects when questioned. The final video was edited to a standard comparable to a Kickstarter or academic demonstration.

Overall Assessment
Our prototype, documentation, and presentation collectively meet or exceed the High Distinction criteria. The project showcases an end-to-end design process that moves from concept to a functioning system with a strong user experience and technical execution. With minor refinements, this prototype could evolve into a publishable or market-ready product.

Below is the detailed checklist for each score band with respective evidences.

Band	Criteria	Question / Criterion	Y/N	Evidence / Notes
PASS	Prototype	Does the prototype include a microcontroller?	Y	ESP32-E central MCU; Sections 2.1-2.2
		Does it contain at least two sensors?	Y	GPS NEO-6M and MMC5603 magnetometer; Section 2.2.1
		Does it provide at least one form of user feedback?	Y	NeoPixel LED ring guidance; Sections 1.2.2, 2.3.5
		Does it include an enclosure?	Y	Custom 3D-printed multi-part enclosure; Section 2.4
		Does it plausibly fit the brief?	Y	Context-aware 'Gandalf staff' navigation device; Sections 1.1, 1.2
	Documentation	Is there a minimal description of the design process?	Y	Process captured across; Sections 1-3
		Is there a full list of components used?	Y	Table 1 in Section 2.2.1
		Are circuit diagrams provided?	Y	Circuit connections described; Fig. 1 referenced in Section 2.2.2
		Was the code provided?	Y	Setup/build steps and repo workflow; Step 2 in Appendix C
		Are challenges described?	Y	Integration issues and trade-offs; Sections 2.2.5, 2.5.5
		Is the motivation for the project described?	Y	Design goal and rationale; Section 1.1
		Was a design rationale provided?	Y	Design rationale; Sections 1.1 and 1.2
		Are similar projects listed?	Y	Similar project is discussed and evaluated; Appendix E
		Is there a user manual?	Y	Extensive User Manual; Appendix D
	Presentation	Was the work presented at the workshop?	Y	Presented on 7 Nov 2025 and worked smoothly
		Was the work presented at the showcase?	Y	Presented on 7 Nov 2025 and worked smoothly
		Was a demo video submitted?	Y	Video prepared and submitted on Canvas
	Overall	If any answer is "No", is there justification or tutor approval?	Y	The project meets all criteria
CREDIT	Prototype	Is sensor data used to infer user context?	Y	Weather + time via APIs; GPS + magnetometer fused; Sections 2.3.4, 2.5
		Does the prototype adapt to user context?	Y	Destination types adapt to weather/time; Section 2.3.4
		Does it use an AI API?	Y	Gemini used to filter places; Sections 2.3.1-2.3.4
		Is there a custom-built enclosure (3D print/laser cut)?	Y	Custom 3D-printed enclosure, multi-part; Section 2.4

		Do all functions work as intended?	Y	Integration/field tests successful; Section 2.5.2, 2.5.3
		Does the idea simulate a sci-fi/fantasy technology?	Y	Gandalf's staff reinterpreted; Sections 1.1
	Documentation	Was the design process thoroughly documented?	Y	Comprehensive report; Sections 1-3
		Is the tutorial clear enough to replicate?	Y	Clear step-by-step tutorial with checklists and photos; Appendix C
		Is the list of similar projects complete?	Y	Similar project is discussed and evaluated; Appendix E
		Is the user manual clear for inexperienced users?	Y	Extensive User Manual; Appendix D
	Presentation	Did the prototype mostly work during presentation?	Y	Presented on 7 Nov 2025 and worked smoothly
		Were most progress reports on time?	Y	Reached all milestones on time; Appendix B and Progress Report
		Did the team narrate the video themselves?	Y	Voice over was narrated by our team member (Sydney). Video submission on Canvas
	Overall	If any answer is "No", is there justification for a credit?	Y	The project meets all criteria
DISTINCTION	Prototype	Is the prototype functional enough for target users?	Y	Prototype passed real tests for target users. Section 2.5.3
		Was the enclosure fully designed by the team?	Y	Enclosure was fully designed iteratively by our team member (Ford). Section 2.4
		Is it novel compared to existing solutions?	Y	The product is innovated from fantasy-inspired context-aware staff. Section 1.1
	Documentation	Is the documentation polished (illustrations, layout, clarity)?	Y	Document is polished and professional, with figures, tables, structured sections; Sections 1-3
		Is the challenge grounded in literature?	Y	It is grounded in The Staff of Gandalf the Wizard from The Lord of the Rings
		Does the process show mastery of design methodology?	Y	The design methodology in this project was splitted into Research & Analysis, Ideation & Concept Development, Design & Prototyping, Testing & Integration and Reflection; Sections 2.5, 3.2
	Presentation	Did all functions work during presentation?	Y	Presented on 7 Nov 2025 and worked smoothly
		Could the team discuss the design process well?	Y	The team has a clear understanding of the design process and a well-defined division of labour; Appendix B
		Could the team answer technical questions?	Y	Technical questions are answered fully and clearly on 7 Nov 2025. The report is comprehensive.
		Were all progress reports on time?	Y	Reached all milestones on time; Appendix B and Progress Report
	Overall	If any answer is "No", is there justification for a distinction?	Y	The project meets all criteria

HIGH DISTINCTION	Prototype	Does it provide a novel and elegant response to the brief?	Y	Creative staff metaphor with robust system; Sections 1.1, 3.3
		Are components used creatively and innovatively?	Y	LED+GPS+Magnetometer navigation in economical way, API fusion; Section 2.3.2-2.3.5
		Is the enclosure designed with user needs in mind?	Y	The enclosure balances ergonomics, ease of assembly, and maintainability; Section 2.4
	Documentation	Is the design rationale creative and inspiring?	Y	Gandalf staff is reinterpreted via pervasive computing with real functionality; Section 1.1, 1.2
		Did the team seek continuous feedback from staff?	Y	Took feedback from tutor and team members during the designing, prototyping and testing stages to iterative on the design
		Were milestones delivered on time?	Y	All milestones are on time including enclosure, wiring, logic&code development and integration; Project Proposal, Project Update, and all deliverables on Canvas
		Did the team display excellent teamwork?	Y	The team has clear roles and collaboration; Appendix B, 3.2.5
	Presentation	Was the presentation polished?	Y	Presentation is polished and professional; Presentation submission on Canvas
		Were creative aids used (props, posters, etc.)?	Y	Prototype, figures, photos, video, and slides used; Video submission on Canvas
		Was the video at Kickstarter/conference standard?	Y	Claimed comparable quality which meets Kickstarter/conference standard. Video submission on Canvas
	Overall	If any answer is “No”, is there justification for HD?	Y	The project meets all criteria

Declaration

Kiba Lin	Ford Pariyavatkul	Yu Shen Qiu	Sydney Jingting Foo
Kiba	***Ford***	***Daniel***	***Sydney***

Date: 6 November 2025

Appendix B: Task Division Summary

Our team divided the work based on each member's strengths while keeping collaboration at the core. Everyone took on both individual and shared responsibilities to make sure the project stayed on track and that all parts of the system came together smoothly.

All Members:

- Responsibilities: Managed electronic components, wiring, and hardware setup.
- Output: Worked together during ideation, prototype trials, report writing, and final testing.

Kiba Lin

- Responsibilities: Worked on software development and GPS module setup. Also handled the purchase of the rake handle.
- Output: Developed GPS functionality and related code, and tested module accuracy

Ford Pariyavatkul

- Responsibilities: Hardware integration, enclosure design, and early code structure.
- Output: Designed and printed the 3D enclosure, refined component fitting, handled assembly planning and execution, and set up the GitHub repository for shared development.

Yu Shen Qiu

- Responsibilities: Code for API interaction and button logic, oversaw code integration.
- Output: Created the bill of materials (BOM), developed API functionality and interaction with API with the button, coordinated the coding integration

Sydney Jingting Foo

- Responsibilities: Coordinated the project timeline, organised meetings, and kept track of milestones.
- Output: Managed scheduling and documentation, guided the team through key decision points, and ensured alignment across software, hardware, and design tasks.

Collaboration Process

We worked in an iterative cycle, meeting weekly to discuss progress, identify issues, and reassign tasks as needed. Each member owned a main area of the project but contributed across others when integration or testing required extra support. Using GitHub for code management and documentation helped keep the workflow transparent and coordinated.

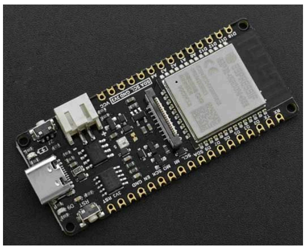
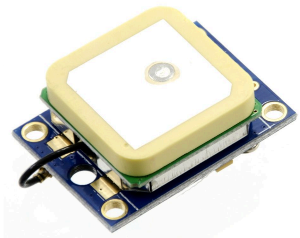
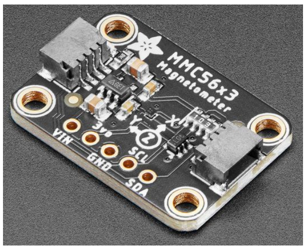
This approach allowed everyone to focus on their strengths while learning from each other, which made the final prototype both functional and cohesive.

Appendix C: Assembly Instructions

Step 1: Prepare Materials

Ensure that all the hardware and enclosure components are ready.
For ease, below is the *checklist* of all the parts, which include:

- ☐ **Hardware**

☐ ESP32-E Microcontroller 	☐ NEO-6 GPS Module 
☐ MMC5603 Magnetometer 	☐ NeoPixel Ring 
☐ 90 Degree USB-C Adaptor 	☐ Rake Handle 1.5m 
☐ Battery 	☐ Mini push button switch 

- [ ] **Enclosure**: There are 8 separate pieces, which we discussed in Section 2.4. Ensure we've all the parts ready before going into the next step.

- [ ] **Tools**: Wires, super glue, clear tape and brown permanent pen (Optional).

Step 2: Software Installation

1. Clone the project repository to your local machine using git clone
   `git clone https://github.com/yushen-qiu/Pervasive_Computing.git`

2. Open VSCode and make sure the PlatformIO extension is installed. Open the cloned project in VSCode, specifically navigating to the 01_Code/firmware folder inside the cloned repository

3. Generate the required API keys. For Weather API, sign up at https://www.weatherapi.com and create a new API key. For Google, go to https://console.cloud.google.com/apis/credentials, create a new API key, and restrict it to only the Places API and Generative Language API.

4. Set up the credentials by navigating to the include/config folder in the firmware folder. Make a copy of Secrets.example.h and rename it to Secrets.h. Open Secrets.h and fill in your Wi-Fi SSID, Wi-Fi password, Google API key, and Weather API key (the program waits until Wi-Fi is connected, so it is good to use a network you can easily turn on and off, such as your phone's mobile hotspot).

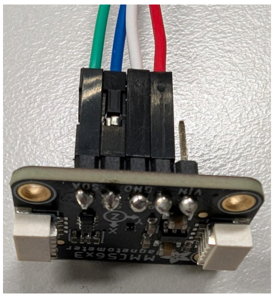
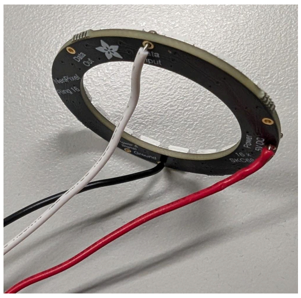
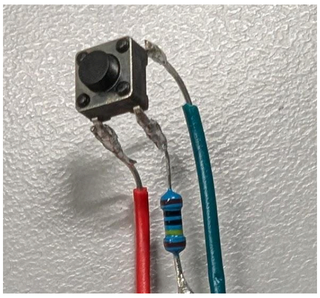
5. Connect your ESP32-E board to your laptop using a USB cable that supports data transfer.

6. Build the project by clicking the Build button (✓ icon) in PlatformIO.

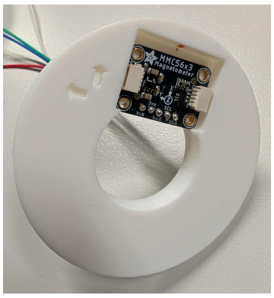
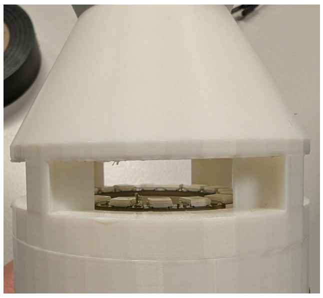
7. Upload the compiled code to the ESP32-E by clicking the Upload button (→ icon).

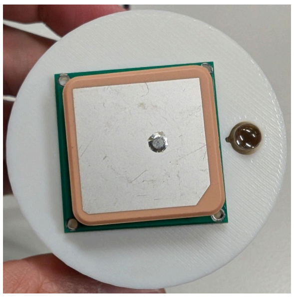
Step 3: Pre-assembly the components

- ☐ Solder all the required connections, and apply electrical tape to secure them further and prevent short circuits.

☐ Solder the magnetometer 	☐ Solder the NeoPixel 
☐ Solder the push button switch 	

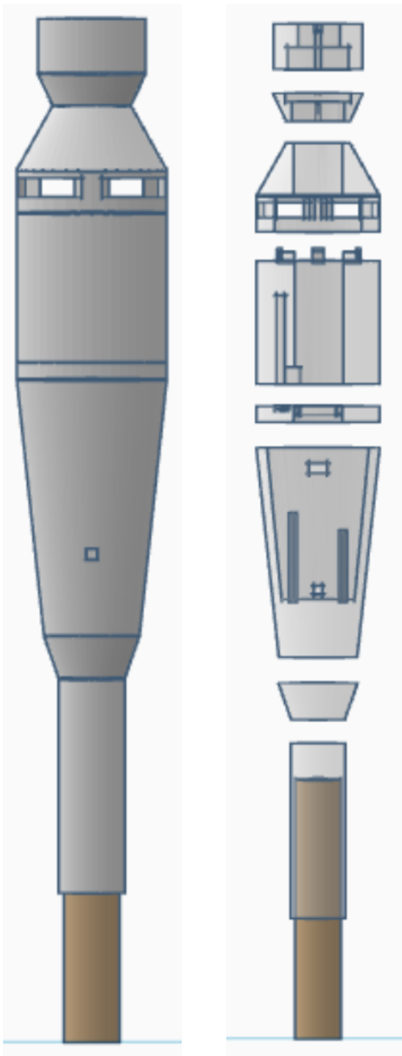
- ☐ Put each component in its respective enclosure part, and apply super glue.

☐ Magnetometer in Mag 	☐ NeoPixel Ring on Neo 

☐ NEO-6 on Top	☐ ESP32-E on Central
	

Step 4: Final Assembling

Connect all the parts using either super glue or clear tape. Between the Central and Mag components, you may prefer clear tape for easier removal. The other parts can be connected with super glue. Apply the glue and wait a few minutes for it to fully cure. The final assembly is shown below.



Appendix D: User Manual

Overview

The Gandalf Staff is an interactive exploration device that guides the user toward nearby destinations. It responds to button input and uses colored LED feedback to indicate interaction phases and navigation directions. The system operates automatically after power-on, requiring no additional setup.

Tips

- For optimal GPS accuracy, use it outdoors with a clear sky view.
- Use a mobile phone for a hotspot WiFi connection to the staff

How to Use

Step 1: Power On
Pick up the staff. Turn on your mobile hotspot, and the staff should connect and light up green

Step 2: Calibrate the Magnetometer
The green light indicates the period of the magnetometer calibration. Please do one rotation while holding the staff, and do a figure eight motion. The staff will light yellow once this has ended.

Step 3: Input Desired Distance

- The staff should now be lit yellow and waiting for button input
- Press the button to activate the 5-second input period
- During the 5-second input window, press the button to set the search radius, each button press adds 200 m. i.e. 4 button presses in the 5-second windows is a 800m search radius
- After the 5-second window is over the staff should be lit red

Step 4: Processing
If the lights are red and blinking, this indicates that the GPS cannot get a reading and that you should move outdoors. Otherwise, after a short pause, the LEDs glow white while the system calculates the destination based on your current GPS position and total button presses.

Step 5: Navigation

- When the LEDs turn blue, the staff begins guiding you.
- The LED ring lights up in the direction you should move toward.
- As you turn, the illuminated LED shifts to stay aligned with your target heading.

Step 6: Arrival
When you are within 100m of the destination, the LED will flash yellow indicating you are getting close. The LED ring brightens fully in a rainbow colour to signal completion. The ESP32 will then be reset with the lights off, waiting to connect to WiFi again.

Appendix E: Review of a Related Project

Project Title: *Gandalf's Magic Staff* by Aaron Haras (Instructables)
Link: https://www.instructables.com/Gandalfs-Magic-Staff/

Overview

This project describes the construction of a wizard-staff prop embedded with lighting, sound, and electronic controls. Key elements include a large crystal element, NeoPixel lighting, hidden switches (via magnets or reed switches), battery and charging circuitry, and optional Bluetooth sound streaming. The build is furnished with a detailed parts list, construction steps (including staff selection, electronics, software patterns), and multiple variations.

Relevance to Our Work

- Like our design, this project uses a microcontroller (Arduino-compatible) to drive LED lighting and user feedback via hidden switches.
- The idea of embedding electronics within a "staff" form factor with context-aware functionality is common to both builds.
- The use of a visually striking "crystal" or glowing top component to provide user feedback resonates with our NeoPixel ring and magnetometer/guidance concept.
- Both projects integrate the aesthetic of a "fantasy wand/staff" with embedded electronics to deliver an interactive experience.

Distinctions & Adaptations

While similar to this example, our project differs substantially in the following ways:

- Our prototype is not merely a cosplay prop but a context-aware navigation tool tailored for orientation and way-finding by leveraging GPS, magnetometer calibration, context adaptation (such as weather/time), and user-direction feedback, rather than purely decorative or theatrical lighting.
- The electronic architecture diverges: our microcontroller logic is firmware-based, supports sensor fusion (magnetometer, GPS, declination, heading) and user interface feedback via the LED ring and haptic/visual cues, rather than relying on NeoPixel lighting and sound patterns alone.
- The enclosure and mechanical design are crafted for field usability, ergonomics, and durability in outdoor navigation contexts, rather than indoor prop aesthetics.
- We have redesigned the flow and user-interaction model to support destination guidance, waypoint logic, and user calibration, which is beyond the typical "magic staff" idea of hidden switches triggering light/sound sequences.

Lessons Learned & Implementation in Our Design

From reviewing the example project we adopted the following useful practices:

1. Hiding electronics within a staff form factor increases user engagemen, so we planned internal PCB layout, battery compartment access and ergonomics early.
2. A comprehensive parts list and step-by-step documentation are crucial for replicability, so we ensured our report and appendices include wiring diagrams, calibration routine and code repository link.

Summary

Including this similar project in our review ensures we satisfy the rubric requirement for discussing alternative/related work. We clearly document how our design is inspired by, but materially different from, the Instructables build. By reflecting on what was done before and how we adapted it for our context-aware navigation brief, we demonstrate awareness of prior art and our project's unique contribution.